MIL-P-46403D (AR)
21 December 1978
SUPERSEDING
MIL-P-46403C (PA)
23 April 1975

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M1B1A2
METAL PARTS FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M1B1A2 for use with blank ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078-Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARD

MILITARY

MIL-STD-1169-Packaging, Packing and Marking for shipment of inert Ammunition Component.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

8838081 - Head
9298013 - Body
8838012 - Cup, Battery
8838015 - Plug, Firing

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.- The metal parts shall comply with all requirements specified on drawings 8838081, 9298013, 8838012 and 8838015 as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.- The primer tubing shall withstand the application of a minimum hydrostatic pressure without rupture or indication of flaws as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.5.2 (see 6.5).

3.4 First article inspection.-This specifcation makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.5 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.5.1 Surface finish.-The requirements for surface finish shall be as detailed on the applicable drawing.

3.5.2 Threads.-Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings.

3.5.3 Burr.-No part shall have a burr which might interfere with the assembling of the primer or which might be injurious to personnel handling the item.

3.5.4 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavities of the primer body and head to insure that no grit, scale, or other foreign matter remains after final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provision of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part description	Drawings	Quantity
Head	8838081	10
Cup Battery	8838012	10
Plug Firing	8838015	10
Body	9298013	10

4.3.2 Inspection to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Primer Percussion M1B1A2		SHEET 1 OF 1		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, battery (Dwg 8838012) Examination for defects	10		3.2	4.4.2.1
	Plug, Firing (Dwg 8838015) Examination for defects	10		3.2	4.4.2.2
	Head (Dwg 8838081) Examination for defects	10		3.2	4.4.2.3
	Body (Dwg 9298013) Examination for defects	10		3.2	4.4.2.4

NOTES:

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4.4 Quality conformance inspectin

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a) Sampling plans.-Unless otherwise specified in the classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46403D (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Cup, Battery	1	1	8838012
				NEXT HIGHER ASSEMBLY
				8839474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Outside diameter, min		0.40%	3.2	Gage
102	Thickness through bottom, max		0.40%	3.2	Gage
103	Length		0.40%	3.2	Gage
Minor					
201	Diameter of hole		0.65%	3.2	Gage
202	Thickness through bottom, min		0.65%	3.2	Gage
203	Inside diameter		0.65%	3.2	Gage
204	Outside diameter, max		0.65%	3.2	Gage
205	Flat missing form surface at open end		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Plug, Firing	1 OF 1	6838015
			NEXT HIGHER ASSEMBLY: 6839474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Large diameter, min		0.40%	3.2	Gage
102	Length and profile of point		0.40%	3.2	Gage
103	Length of pilot, min		0.40%	3.2	Gage
104	Total length		0.40%	3.2	
Minor					
201	Large diameter, max		0.65%	3.2	Gage
202	Length of pilot, max		0.65%	3.2	Gage
203	Diameter of pilot		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.5	

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46403D (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.3	Head	1 2	8838081
			NEXT HIGHER ASSEMBLY 8839474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Pitch diameter of body thread		0.40%	3.2	Gage
102	Major diameter of body thread, min		0.40%	3.2	Gage
103	Diameter of body		0.40%	3.2	Gage
104	Diameter of battery cup cavity		0.40%	3.2	Gage
105	Diameter of percussion element cavity		0.40%	3.2	Gage
106	Depth of percussion element cavity		0.40%	3.2	Gage
107	Thickness of flange, max		0.40%	3.2	Gage
108	Crack or split		0.40%	3.2	Gage
109	Flash hole missing or obstructed		0.40%	3.2	Visual
Minor					
201	Major diameter of body thread, max		0.65%	3.2	Gage
202	Diameter of flange		0.65%	3.2	Gage
203	Depth of battery cup cavity		0.65%	3.2	Gage
204	Concentricity between percussion element cavity and flash hole		0.65%	3.2	Gage
205	Concentricity between body diameter and percussion element		0.65%	3.2	Gage
206	Concentricity between body diameter and flange		0.65%	3.2	Gage
207	Diameter of chamfer at mouth of Percussion element cavity		0.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46403D (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Head	2	2	8838061
				NEXT HIGHER ASSEMBLY: 8639474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
208	Diameter of flash hole		0.65%	3.2	Gage
209	Thickness of flange, min		0.65%	3.2	Gage
210	Length to shoulder above body thread		0.65%	3.2	Gage
211	Length of thread undercut, max		0.65%	3.2	Gage
212	Total length		0.65%	3.2	Gage
213	Radius missing		0.65%	3.2	Visual
214	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46403D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Body	1 OF 2	9296013
			NEXT HIGHER ASSEMBLY 8839474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Flash hole missing		100%	3.2	Visual
2	Crack or split		100%	3.2	Visual
Major					
101	Hydrostatic test (see note)		100%	3.3	4.5.2
102	Pitch diameter of thread		0.40%	3.2	Gage
103	Length of thread, min		0.40%	3.2	Gage
104	Minor diameter of thread, max		0.40%	3.2	Gage
105	Wall thickness, min		0.40%	3.2	Gage
Minor					
201	Length of thread, max		0.65%	3.2	Gage
202	True position of inside diameter		0.65%	3.2	Gage
203	Thickness at closed end		0.65%	3.2	Gage
204	Location of first row of holes		0.65%	3.2	Gage
205	Inside diameter, min		0.65%	3.2	Gage
206	Depth of counterbore, max		0.65%	3.2	Gage
207	Diameter of counterbore		0.65%	3.2	Gage
208	Total length, max		0.65%	3.2	Gage
209	Plug gage fails to enter body to required depth		0.65%	3.2	Gage

NOTES:

If any primer tubing fails to comply with the applicable requirement it shall be rejected. No primer bodies shall be fabricated from rejected tubing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.4	Body	2 2	9298013
			NEXT HIGHER ASSEMBLY 8839474

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Body fails to pass through required ring gage		0.65%	3.2	Gage
211	Hole in swaged end		0.65%	3.2	Visual
212	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection Tables

4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 - herein.

4.5 Method of Inspection

4.5.1 Materials, components and processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specififed manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at a minimum hydrostatic pressure as calculated by the following formula.

$$P = \frac{2\ St}{D} \quad X \quad .9$$

where:

P= test pressure in pounds per square inch (psi).
S = minimum specified yield strength, psi.
t = specified minimum wall thickness in inches.
D = specified maximum outside diameter in inches.

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Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PACKAGING

5.1 Packing and marking.-Packing and marking shall be in accordance with MIL-STD-1169 Level B.

6. NOTES

6.1 Intended use.- The components covered by this specification are intended for use on 105mm, 90mm and 75mm blank ammunition.

6.2 Ordering data.- See MIL-A-48078.

6.3 Submission of inspection equipment for design approval.-See MIL-A-48078. Submit designs as required to Commander US Army Armament Research and Development Command ATTN: DRDAR-QAR-I Dover, NJ 07801.

6.4 Drawings.-Drawings listed in section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings pepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Hydrostatic test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

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